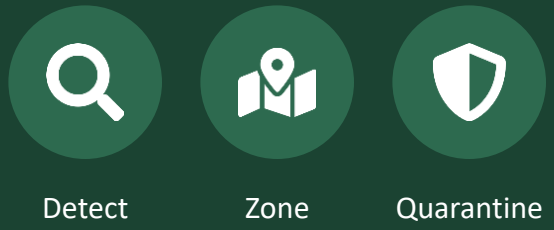
AI-Based Crop Disease Detection, Quarantine and Zoning System

Early disease detection, risk zoning and quarantine advice. Supports SDG 2.3



Introduction & Problem

Healthy crops feed people
Food security and farmer income depend on them.

Diseases spread fast
Fungi, bacteria and viruses can reach whole areas.

Early detection is hard
Manual checks and expert help are limited.

Problem Statement

No single system finds the disease, estimates risk zones and advises quarantine.

Our Solution

AI reads leaf images, classifies risk zones and suggests preventive steps.

Objectives & Requirements

- Identify crop diseases using AI
- Analyze leaf images with ML / DL
- Understand how diseases spread
- Find areas with different risk levels
- Give quarantine recommendations
- Generate alerts about disease risk
- Support early detection and prevention
- Support sustainable farming (SDG 2.3)

Software

- Python, VS Code / Colab
- TensorFlow or PyTorch, OpenCV
- Scikit-learn, Streamlit

Hardware & Data

- Computer, minimum 4 GB RAM
- Internet connection
- Healthy and diseased leaf images

Crop Diseases & How They Spread

Common diseases

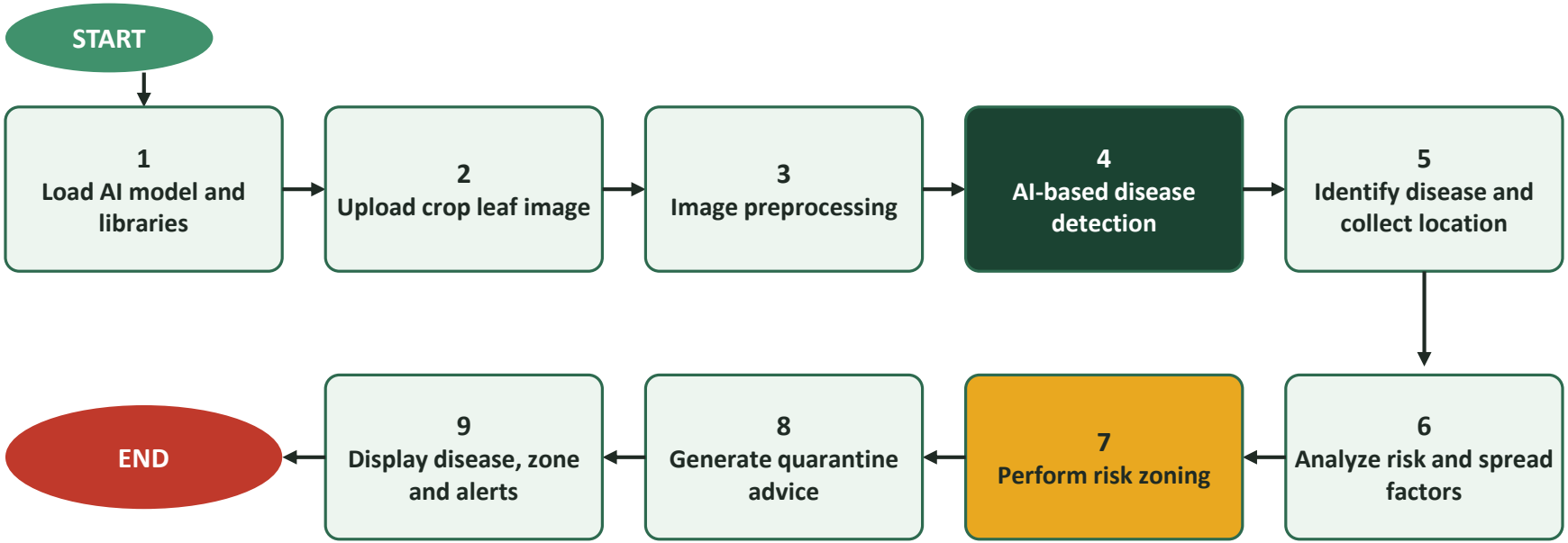
- **Leaf Blight:** Damages leaf tissue
- **Leaf Rust:** Rust-like spots on leaves
- **Powdery Mildew:** White powder-like growth
- **Bacterial Leaf Spot:** Spots caused by bacteria
- **Mosaic Disease:** Irregular patterns and discoloration

Spread methods

- Wind
- Water
- Soil
- Insects
- Infected seeds
- Dirty tools
- People and vehicles

Know the route to pick prevention

Algorithm & Flowchart



Flow: leaf image → disease → risk zone → quarantine advice

Methodology: System Modules

1. Image Input
Leaf image from user

2. Processing
Resize and normalize

3. Detection
CNN predicts disease

4. Location
Field or area details

5. Risk Analysis
Reports and spread factors

6. Zoning
Low, medium or high risk

7. Quarantine
Preventive advice

8. Output
Disease, zone, advice

Risk zones: Low-risk zone | Medium-risk zone | High-risk zone

Result: Sample Outputs

Chilli: CLCV
Confidence: 78%
Location: Mamidada

HIGH-RISK ZONE

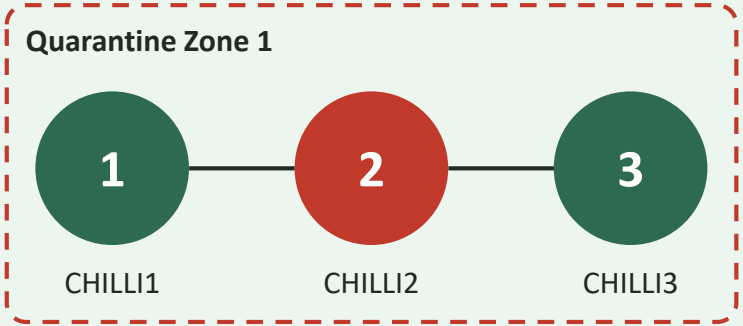
Isolate farms, restrict movement, inform authorities.

Paddy: Brown spot
Confidence: 30%
Location: Rajanagaram

LOW-RISK ZONE

Monitor crops, keep fields clean, use healthy planting material.

Farm network (graph analysis)



- CHILLI2 is the critical farm
- It connects the other two farms
- Impact: CHILLI2 = 2, others = 1

Conclusion & Future Scope

Conclusion

- AI finds possible crop diseases from leaf images.
- Reports and location give risk zones and quarantine advice.
- A prototype: it supports, not replaces, experts.

Supports SDG 2.3

Future Enhancements

- Real-time detection
- GPS-based zoning
- Weather data
- Advanced deep learning
- Mobile app
- Multilingual support
- Automated alerts
- Expert consultation
- Live field monitoring
- Risk prediction